Haitamim Jahran Mahendra

AI Engineer

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Professional Summary

An Informatics Engineering student at Universitas Islam Negeri Syarif Hidayatullah Jakarta with a passionate drive to develop practical skills in Data Engineering and AI automation. Possesses hands-on, project-based experience in building data ingestion flows, text normalization scripts, and localized Retrieval-Augmented Generation (RAG) applications using Python and LangChain (LCEL). An adaptable and enthusiastic learner who is committed to supporting engineering teams, mastering industry-standard data platform workflows, and growing under professional mentorship.

Technical Project & Experience

Waguri AI – Personal Project |July 2026

Full-Stack AI & Data Engineering Developer | Tangerang, Indonesia

- The application required an interactive system capable of handling unstructured developer portfolio quieres with minimal latency
- Design and deploy a full-stack Retrieval-Augmented Generation (RAG) conversational agent capable of automated context extraction.
- Architected the system framework using LangChain Expression Language (LCEL) and Streamlit, built an automated ETL ingestion pipeline using TextLoader and RecursiveCharacterTextSplitter, generated semantic embeddings via HuggingFace (all-MiniLM-L6-v2), and indexed the processed data into a FAISS vector database while setting up prompt engineering guardrails against injections. Integrated real-time Text-to-Speech using gTTS and BytesIO.
- Delivered a stable, secure production pipeline achieving sub-second semantic search responses with zero execution runtime crashes.

Artificial Intelligence Literacy and Innovation Institute (ALII) UIN Jakarta | April 2026 – June 2026

Data Engineer Intern | Jakarta, Indonesia

- High-volume multi-book Islamic text databases (Kutubus Sittah) distributed across a massive 1GB unstructured CSV dataset lacked direct relational mappings to their corresponding Indonesian translation SQL databases, resulting in critical data isolation.
- Engineer a high-performance data extraction, text normalization, and semantic matching alignment pipeline to pair thousands of original Arabic hadith strings with correct translations, followed by an automated structured Markdown wiki generation lifecycle.
- Developed a memory-optimized chunking ingestion script using Python (Pandas) to process the 1GB data array, constructed an Arabic text normalization helper

algorithm to strip diacritics (harakat/tashkeel) while strictly maintaining core XML indexing tags (<SANAD>, <NAR>, <MATN>), deployed a high-accuracy dual-track matching matrix combining TF-IDF vectorization with K-Nearest Neighbors (KNN) Cosine Similarity, and orchestrated automated frontmatter meta-summary refinements using the Qwen 3 4B-Instruct large language model.

- Successfully matched and consolidated 5,348 unstructured text entities into an integrated data repository, achieving a 97.46% average semantic alignment accuracy score (98.53% median score) and completing 100% of the automated Markdown summary refinement phase.

Machine Learning Project – Dicoding Certification | April 2026
Data Analyst & Predictive Modeler | Jakarta, Indonesia (Remote)

- High-volume customer transaction profiles contained hidden behavioral segments and unorganized financial risk indicators.
- Identify distinct customer behavioral groups and build an automated framework to predict target client classifications based on historical records.
- Engineered advanced data clustering structures utilizing the K-Means Clustering algorithm for behavioral segmentation, and trained robust classification systems via Decision Tree and Random Forest architectures alongside meticulous hyperparameter tuning.
- Significantly enhanced the clarity of target demographic segmentation and optimized overall classification scoring matrices for predictive data analytics.

Movie Recommender System – Research Project | February 2026
Data Science Researcher | Tangerang, Indonesia

- Digital media applications frequently suffer from navigation bottlenecks and user "Choice Paralysis" due to rigid text search indexing workflows.
- Build an automated semantic preference mapping module to generate accurate, context-aware user suggestions from massive text databases.
- Deployed Content-Based Filtering algorithms by executing text tokenization logic through TF-IDF arrays and evaluating algorithmic proximity using Cosine Similarity scores.
- Designed an active automated suggestion module that successfully delivered contextually precise and highly relevant recommendations.

Cyrptographic CLI Tool – Personal Project | June 2025
Python Backend Developer | Tangerang, Indonesia

- Localized text data inputs required automated syntax validation rules and secure encryption matrices without depending on heavy third-party software overhead.
- Develop a resilient, error-resistant command-line terminal utility for localized text encryption and data validation routines.
- Formulated a CLI-based module in Python using modular arithmetic, targeted string parsing loops, and custom try-except structured exception handling frameworks.
- Produced a stable data utility ensuring 100% data input sanitization rates and absolute logical execution error prevention.

Web Design Experiments – Self Learning | May 2025
Frontend Developer | Tangerang, Indonesia

- Needed to bridge the functional gap between backend raw data structures and

interactive user-facing web interfaces.

- Construct semantic and responsive static pages to thoroughly master visual document layout structures and styling patterns.
- Coded multi-device static web pages utilizing HTML5 semantic tags and CSS3 layout layouting rules to understand DOM manipulation boundaries.
- Built clean, reusable, and semantic modular code serving as a strong layout foundation for future full-stack integrations.

Education

Universitas Islam Negeri Syarif Hidayatullah Jakarta | August 2023 – Present
Bachelor of Engineering (S.T.) in Informatics Engineering (Six Semester Student) | Jakarta, Indonesia

- **Academic Standings:** Current Cumulative GPA: 3.48 / 4.00.
- **Core Coursework:** Algorithm & Data Structures, Advanced Programming Logic, Artificial Intelligence Basics, Database & Web Engineering.

Madrasah Aliyah Negeri 10 Jakarta Barat | 2020-2023
High School Diploma majoring in Social Sciences (IPS) | Jakarta, Indonesia

Skills

- **Hard Skills:** Data Engineering Core, ETL Pipelines, Ingestion Workflows, RAG Architecture, Text Normalization Algorithms, Dual Matching Systems (TF-IDF + KNN), Text Similarity Metrics, Semantic Search.
- **Programming Languages:** Python (Advanced Logic, Data Structures, LCEL, Pandas, Scikit-Learn), SQL (Basic Data Queries), HTML5, CSS3.
- **Tools & Technologies:** LangChain Framework, Qwen LLM, FAISS Vector DB, Streamlit Dashboarding, Git Version Control, Linux Shell & Basic Automation, HuggingFace Transformers, PyCharm, VS Code.
- **Soft Skills:** Algorithmic Thinking, Fast Technology Adaptation, Analytical Problem Solving, Technical Documentation, Professional Team Communication.

Certifications

- **Belajar Machine Learning untuk Pemula** – Issued by Dicoding Academy (April 2026)
- **Belajar Dasar AI** – Issued by Dicoding Academy (February 2026)
- **Memulai Pemrograman dengan Python** – Issued by Dicoding Academy (March 2026)
- **English Language Proficiency Certificate** – Center for Language Development UIN Jakarta

Portofolio

GitHub Repository:

https://github.com/PacarKaoruko

Waguri AI Project: https://waguri-ai-imikerenaja.streamlit.app/